AI Product & Intelligent Systems

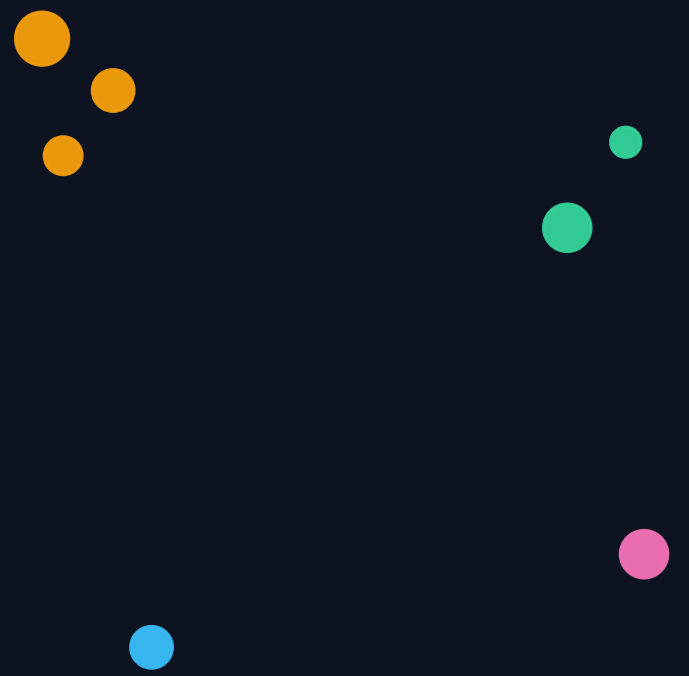
TasteBridge

User-Controlled Music Discovery

A music discovery tool that recommends tracks from either seed songs or a natural-language description, while letting users control how familiar or adventurous the results should be.

Business problem

Current recommenders can feel repetitive and opaque. Users have limited control over how far discovery moves beyond their usual listening habits.



Visualises taste space: close, bridge, frontier

User journey

From taste or intent to controllable discovery.

Input A
Seed tracks

Input B
Text description

→

Embeddings
Audio + text vectors

→

Ranking
Relevance + distance

→

Results
Safe / Bridge / Wildcard



Taste Bridge visual
Shows why an unexpected track connects back to the user's starting point.

Core differentiator: the user controls exploration instead of passively receiving “more of the same.”

Data science approach

A practical MVP using open data and pretrained models.

Data sources

Free Music Archive
MusicBrainz
ListenBrainz (stretch)

Tech stack

Python, Jupyter, pandas, NumPy, scikit-learn, PyTorch, Hugging Face, CLAP, librosa, FAISS, NetworkX, Plotly, Streamlit, Git, GitHub

Methodology

EDA, preprocessing, feature engineering, audio/text embeddings, cosine similarity, nearest-neighbour retrieval, clustering, custom ranking, novelty/diversity scoring, recommender evaluation.

Primary ML object

Embeddings

Recommendation logic

Ranking

Product layer

Streamlit

Data availability

High

MVP principle: do not train a large audio model. Use existing embeddings and focus the capstone on controllable discovery and evaluation.

Four-week MVP scope

Build the smallest version that proves the idea.

Week 1
Data + baseline
Track → similar tracks

→

Week 2
Discovery logic
Safe / Bridge / Wildcard

→

Week 3
Intent + visual
Text input + Taste Bridge

→

Week 4
Evaluation + polish
Demo, README, slides

Recommended team size: 3 people

1. ML/recommendation logic
2. Data + evaluation
3. Product demo + visualisation

Evaluation question

Can users get more novel and interesting music recommendations while still finding them relevant?

Final deliverables

Working demo app, recommendation pipeline, evaluation results, project README and presentation.

Scope rule: collaborative filtering and Spotify integration stay as stretch goals